

Tutorial Sheet - 03

PART - A

Q.1 Define order and degree of a differential equation.

Solⁿ Order :- The order of a differential equation is the highest order of derivative present in the equation.

Eg :- $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + y = 0$, here, the highest derivative is $\frac{d^2y}{dx^2}$

So, the order is 2.

Degree :- The degree of a differential equation is the highest power of the highest order derivative, after the equation is made free from radicals and fraction in derivatives.

Eg :- $\left(\frac{d^2y}{dx^2}\right)^3 + \frac{dy}{dx} + y = 0$. Here, the highest order derivative is second order and its power is

3, so the degree is 3.

Q.2 Write the standard form of linear first order differential.

Solⁿ The standard form of a linear first order differential equation is:

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \dots \text{where } P(x) + Q(x) \text{ are function of } x, \\ y \text{ is dependant variable.}$$

Q.3 Write what is Integrating factor? Write the integrating factor for

$$\frac{dy}{dx} + P(x)y = Q(x).$$

solⁿ Integrating Factor (I.F) :- An I.F is a function by which a linear first order differential equation is multiplied to make it an exact equation, so that it can be solved easily.

For exⁿ :- $\frac{dy}{dx} + P(x)y = Q(x)$

I.F is :- $e^{\int P(x)dx}$

Q.4 Define an exact differential equation and state the condition for exactness.

solⁿ Exact Differential Equation :- A differential equation of the form
 $M(x,y)dx + N(x,y)dy = 0$
is said to be exact differential equation.

Condition for Exactness :- The equation is exact if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Q.5 Write the standard form of Bernoulli's differential equation.

solⁿ The standard form of Bernoulli's differential equation is :-

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \quad \dots \text{where } P(x) + Q(x) \text{ are functions of } x,$$

- n is any real number (with $n \neq 0, 1$)
- y is the dependent variable.

PART - B

Q.1 solve the differential equation : $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$ ——— (1)

Solⁿ $M = 2xy + y^2$, $N = x^2 + 2xy$

$$\frac{dM}{dy} = 2x + 2y$$

$$\frac{dN}{dx} = 2x + 2y$$

$$\left\{ \frac{dM}{dy} = \frac{dN}{dx} \right\}$$

So, eqⁿ (1) is exact diff. eqⁿ

Solⁿ :- $\int M dx + \int N dy = C$

$$\int (2xy + y^2) dx + \int (x^2 + 2xy) dy = C$$

$$\Rightarrow x^2 y + xy^2 = C$$

Q.2 solve the differential equation : $\frac{dy}{dx} + y \tan x = \sin x$ ——— (1)

Solⁿ Compare eqⁿ (1) by general eqⁿ $\frac{dy}{dx} + P/y = Q$

$$P = \tan x, \quad Q = \sin x$$

$$I.F = e^{\int \tan x dx} \Rightarrow e^{\log \sec x} \Rightarrow \sec x$$

Solⁿ $\Rightarrow y \cdot (I.F) = \int Q \cdot (I.F) dx + C$

$$y \cdot \sec x = \int \sin x \cdot \sec x dx + C$$

$$y \cdot \sec x = \int \frac{\sin x}{\cos x} dx + C \quad \therefore \{ \sec x = \cos^{-1} x \}$$

$$y \cdot \sec x = \int \tan x dx + C \quad \therefore \{ \int \sin x / \cos x = \tan x \}$$

$$y \cdot \sec x = \log |\sec x| + C$$

$$y = \frac{1}{\sec x} \log |\sec x| + C$$

$$\Rightarrow y = \cos x (\ln |\sec x| + C)$$

Q.3 Solve the Bernoulli's differential equation: $\frac{dy}{dx} + y = y^2$. — (1)

Solⁿ Divide the eqⁿ (1) by y^2 , then

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{y} = 1 \quad \text{--- (2)}$$

Put $\frac{1}{y} = t \Rightarrow y^{-1} = t \Rightarrow -\frac{1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$

$$\frac{dy}{dx} = -y^2 \frac{dt}{dx} \quad \therefore \{ \text{Put in eqⁿ (2)} \}$$

$$\frac{1}{y^2} - \left(-y^2 \frac{dt}{dx} \right) + \frac{1}{y} = 1$$

$$\Rightarrow \frac{dt}{dx} - t = -1 \quad \text{--- (3)}$$

eqⁿ (3) is L.D in t

$$P = -1, \quad Q = -1$$

$$I.F = e^{\int -1 dx} \Rightarrow e^{-x}$$

Complete solⁿ :- $t(I.F) = \int \theta(I.F) dx + C$

$$t \cdot e^{-x} = \int (-1 \cdot e^{-x}) dx + C$$

$$t \cdot e^{-x} = e^{-x} + C$$

$$\frac{1}{y} e^{-x} = e^{-x} + C$$

$\therefore \{ \text{Put value of } t \}$

$$\frac{1}{y} = \frac{e^{-x} + C}{e^{-x}} \Rightarrow y = \frac{e^{-x}}{e^{-x} + C} \quad \{ \text{Take } e^{-x} \text{ common} \}$$

$$\Rightarrow y = \frac{e^{-x}}{e^{-x}(1 + C e^x)}$$

$$\Rightarrow y = \frac{1}{1 + C e^x} \quad \text{Ans}$$

PART - C

Q.1 Solve the differential equation: $(x^2y + y^3)dx + (x^3 + 3xy^2)dy = 0$.

$$\text{Sol}^n \quad \frac{\partial M}{\partial y} = \frac{\partial (x^2y + y^3)}{\partial y} = x^2 + 3y^2, \quad \frac{\partial N}{\partial x} = \frac{\partial (x^3 + 3xy^2)}{\partial x} = 3x^2 + 3y^2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$I.F = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{(x^2 + 3y^2) - (3x^2 + 3y^2)}{x^3 + 3xy^2} = \frac{-2x^2}{x(x^2 + 3y^2)}$$

$$I.F = \frac{1}{Mx + Ny} \Rightarrow Mx + Ny \Rightarrow (x^2y + y^3)x + (x^3 + 3xy^2)y$$

$$\Rightarrow Mx + Ny = x^3y + xy^3 + x^3y + 3xy^3 \Rightarrow 2x^3y + 4xy^3$$

$$(x^2y dx + x^3 dy) + (y^2 dx + 3xy^2) dy = 0$$

$$d(x^2y) = 3x^2y dx + x^3 dy$$

$$d(xy^3) = y^3 dx + 3xy^2 dy$$

Q.2 Solve the Bernoulli's differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2 \cdot y^2$ — (1)

Solⁿ Divide the eqⁿ (1) by y^2 , then

$$\frac{1}{y^2} \frac{dy}{dx} + \frac{1}{xy} = x^2 \quad \text{--- (2)}$$

Put $\frac{1}{y^2} = t \Rightarrow y^{-1} = t \Rightarrow \frac{dy}{dx} = -y^2 \frac{dt}{dx} \therefore$ [Put in eqⁿ (2)]

$$\frac{1}{y} \left(-y^2 \frac{dt}{dx} \right) + \frac{t}{xy} = x^2$$

$$\frac{dt}{dx} - \frac{t}{x} = -x^2 \quad \text{--- (3)}$$

eqⁿ (3) is L.D in t

$$P = -\frac{1}{x}, \quad Q = -x^2$$

$$\text{I.F} = e^{\int \frac{1}{x} dx} \Rightarrow e^{-\log x} \Rightarrow x^{-1}$$

Complete solⁿ = $t(\text{I.F}) = \int Q(\text{I.F}) dx + C$

$$t \cdot x^{-1} = \int (-x^2 \cdot x^{-1}) dx + C$$

$$t \cdot x^{-1} = -\frac{x^2}{2} + C$$

$$t = -\frac{x^2}{2} + C \Rightarrow \frac{1}{y} = -\frac{x^2}{2} + Cx \quad \text{[Put the value of } t \text{]}$$

$$y = \frac{1}{Cx - \frac{x^2}{2}}$$

Q.3 Solve the linear differential equation: $x \frac{dy}{dx} + y = x \cos x$. — (1)

solⁿ compare eqⁿ (1) by general eqⁿ $\frac{dy}{dx} + Py = Q$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \cos x \quad \left\{ \because \text{Dividing both side by } x \right\}$$

$$P = +\frac{1}{x}, \quad Q = \cos x, \quad \text{I.F} = e^{\int \frac{1}{x} dx} = e^{\log x} \Rightarrow x$$

$$\text{Complete Solⁿ :- } y (\text{I.F}) = \int Q (\text{I.F}) dx + C$$

$$y \cdot x = \int x \cdot \cos x \, dx + C$$

$$y \cdot x = x \sin x - \cos x + C \quad \left\{ \text{By integration by parts} \right\}$$

$$y = \sin x - \frac{\cos x}{x} + C$$

$$t = -\frac{x^3}{2} + C \Rightarrow \frac{1}{y} = -\frac{x^3}{2} + Cx \quad \therefore \text{Put the value of } t$$

$$y = \frac{1}{Cx - \frac{x^3}{2}} \quad \text{Ans}$$

Q.3 Solve the linear differential equation : $x \frac{dy}{dx} + y = x \cos x$ — (1)

Solⁿ compare eqⁿ (1) by general eqⁿ $\frac{dy}{dx} + Py = Q$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = \cos x \quad \left\{ \because \text{Dividing both side by } x \right\}$$

$$P = +\frac{1}{x}, \quad Q = \cos x, \quad \text{I.F} = e^{\int \frac{1}{x} dx} = e^{\log x} \Rightarrow x$$

$$\text{Complete Sol}^n :- y (\text{I.F}) = \int Q.(\text{I.F}) dx + C$$

$$y.x = \int x \cos x dx + C$$

$$y.x = x \sin x - \cos x + C \quad \left\{ \text{By integration by parts} \right\}$$

$$y = \sin x - \frac{\cos x}{x} + C \quad \text{Ans}$$

Ok
27/2/22